H3IM FX Swap Arbitrage

Systematic G10 & EM FX Swap Arbitrage Strategy

Track Record: Apr 2024 - Jun 2026 (27 Months / 2.2 Years) | **Benchmark:** S&P 500 Total Return

ATHÉNÉE INVESTMENT

CAPITAL INTRODUCTION & THIRD-PARTY MARKETING

CAGR (Net)	Ann. Volatility	Sharpe Ratio	Max Drawdown	Ann. Alpha (α)
29.08%	**2.06%**	**12.53**	**0.00%**	**26.17%**
S&P 500: 18.64%	S&P 500: 12.67%	S&P 500: 1.42 | Sortino: -	S&P 500: -7.50% | Calmar: -	Beta: -0.02 | Corr: -0.12

Performance, Risk & Hit Rate Indicators

Indicator	Fund	S&P 500
CAGR (Compound Annual Growth)	**29.08%**	18.64%
Annualized Mean Return	25.82%	17.98%
Annualized Volatility (Std. Dev.)	2.06%	12.67%
Proxy Sharpe Ratio ($r_f = 0\%$)	**12.53**	1.42
Annualized Sortino Ratio	-	2.95
Maximum Drawdown (MDD)	**0.00%**	-7.50%
Calmar Ratio (CAGR / |MDD|)	-	2.49
Cumulative Total Return	77.61%	46.91%
Best / Worst Monthly Return	+3.32% / +1.22%	+10.49% / -5.63%
Positive Months (Hit Rate)	100.0% (27/27)	66.7% (18/27)
Crisis Hit Rate (Fund > 0 | SPX < 0)		**100.0% (9/9)**
Up-Mkt Loss Rate (Fund < 0 | SPX > 0)		0.0% (0/18)

Alpha, Beta, Correlation & Tail Analysis (vs. S&P 500)

Quantitative Metric	Estimate / Diagnostic
Annualized Alpha (α) & t-stat	**26.17% (t=17.36)**
Market Beta (β_{SPX}) & t-stat (R^2)	-0.02 (t=-0.60) ($R^2 = 1.4\%$)
Pearson Correlation (r) to S&P 500	**-0.12** (95% Blk CI: [-0.45, 0.14])
Spearman Rank Correlation (ρ)	-0.07 (95% Blk CI: [-0.46, 0.22])
Upside / Downside Capture Ratio	Up: **0.64** | Down: **-0.85**
Conditional Up-Market (SPX > 0)	$\alpha = 29.8\%, \beta = -0.08$
Conditional Down-Market (SPX < 0)	$\alpha = 22.2\%, \beta = -0.09$
Tail Corr. (SPX $<$ 1st Quartile)	$r = -0.25, \rho = -0.43$ ($n = 7$)
Tail Corr. (SPX $<$ 1st Decile)	$r = -0.97, \rho = -1.00$ ($n = 3$)
Worst 20% SPX Months Avg. Return	Fund: +2.22% vs SPX: -3.22%
Cross-Asset Beta (WTI / BTC)	WTI β: -0.00 | BTC β: 0.00
Ljung-Box Autocorrelation	$p_1 = 0.36,\ p_3 = 0.48$ (Pass (No Serial Corr.))

Historical Monthly & Yearly Net Return Matrix (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Yearly
2024	-	-	-	1.4%	1.4%	1.2%	2.4%	3.0%	3.3%	1.4%	2.9%	3.3%	**22.1%**
2025	1.4%	1.9%	2.8%	2.1%	2.3%	1.9%	2.0%	2.0%	2.2%	2.1%	2.6%	3.0%	**29.9%**
2026	2.2%	1.8%	1.9%	1.8%	1.8%	2.0%	-	-	-	-	-	-	**12.0%**

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Capital Introduction & Third-Party Marketing Memorandum. Strictly for Institutional & Qualified Investors. Please see Page 2 for Risk & Legal Disclosures.

Appendix, Quantitative Methodology & Risk Disclosures

Fund Reference: H3IM FX Swap Arbitrage (Apr 2024 - Jun 2026 (27 Months / 2.2 Years))

Appendix A: Extended Statistical, Bootstrap & Factor Diagnostics

Diagnostic Family	Methodology / Statistical Specification	Empirical Value
Pearson Correlation CI	95% Confidence Interval (Standard IID Bootstrap vs. Moving Block $B = 3$, 1,000 iters)	Std: [-0.42, 0.28] | Blk: [-0.45, 0.14]
Spearman Rank Corr. CI	95% Confidence Interval (Rank Monotonic Co-Movement, Standard vs. Block $B = 3$)	Std: [-0.44, 0.35] | Blk: [-0.46, 0.22]
Worst-Case Co-Dependence	97.5th Percentile Upper-Bound Block Bootstrap Correlation (Nearest PSD Projection)	$\rho_{\text{wc}} = 0.14$
Serial Correlation (Lag 1 & 3)	Ljung-Box Q-Statistic p-values testing for return smoothing / illiquidity bias	$p_1 = 0.36,\ p_3 = 0.48$ (Pass (No Serial Corr.))
Conditional Up-Market OLS	Annualized Alpha & Beta estimated strictly during positive S&P 500 months ($n = 18$)	$\alpha = 29.8\%, \beta = -0.08\ (t_\alpha = 10.97)$
Conditional Down-Market OLS	Annualized Alpha & Beta estimated strictly during negative S&P 500 months ($n = 9$)	$\alpha = 22.2\%, \beta = -0.09\ (t_\alpha = 5.11)$
Commodity Factor (WTI)	OLS Regression against Front-Month WTI Crude Oil Monthly Returns	$\beta = -0.00\ (t = -0.47),\ \alpha = 25.8\%$
Digital Asset Factor (BTC)	OLS Regression against Bitcoin (BTC-USD) Monthly Returns	$\beta = 0.00\ (t = 0.46),\ \alpha = 25.8\%$

Appendix B: Quantitative Framework & Indicator Definitions

- **CAGR & Annualized Volatility:** Compound Annual Growth Rate is computed geometrically as $\text{CAGR} = \left(\prod_{t=1}^{T}(1 + r_t)\right)^{12/T} - 1$. Annualized Volatility is sample standard deviation scaled by $\sqrt{12}$: $\sigma_{\text{ann}} = \hat{\sigma}_m\sqrt{12}$. Proxy Sharpe Ratio assumes $r_f = 0\%$ ($\bar{r}_{\text{ann}}/\sigma_{\text{ann}}$).
- **Maximum Drawdown (MDD) & Calmar Ratio:** Maximum peak-to-trough decline of cumulative compounded wealth $W_t = \prod_{s\le t}(1 + r_s)$, defined as $\text{MDD} = \min_t\left((W_t - \max_{s\le t} W_s)/\max_{s\le t} W_s\right)$. Calmar Ratio is CAGR/|MDD|.
- **OLS Alpha (α), Beta (β) & Capture Ratios:** Estimated via Ordinary Least Squares $r_{\text{fund},t} = \alpha_m + \beta r_{\text{SPX},t} + \varepsilon_t$, where $\alpha_{\text{ann}} = 12 \times \alpha_m$. Upside (Downside) Capture measures the ratio of the fund's mean return to the S&P 500's mean return during positive (negative) S&P 500 months.
- **Tail Correlations & Moving Block Bootstrap:** Tail correlations evaluate linear (r) and rank (ρ) co-movement during S&P 500 drawdowns below the 25th percentile (Q_1) and 10th percentile (D_1). Moving Block Bootstrap ($B = 3$ months) preserves local serial dependencies when constructing 95% confidence intervals.

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